

TYGACIL®
(Tigecycline)

1. NAME OF THE MEDICINAL PRODUCT

TYGACIL®

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each 5 mL tigecycline vial contains 50 mg of tigecycline lyophilized powder for intravenous infusion and 100 mg of lactose monohydrate. The pH is adjusted with hydrochloric acid, and if necessary sodium hydroxide. After reconstitution, 1 mL contains 10 mg of tigecycline.

For a full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Tigecycline 50 mg sterile, lyophilized powder for intravenous infusion is an orange lyophilized powder or cake.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Tigecycline is indicated for the treatment of the following infections caused by susceptible strains of the designated microorganisms in the conditions listed below for patients 18 years of age and older:

Complicated skin and skin structure infections caused by *Escherichia coli*, *Enterococcus faecalis* (vancomycin-susceptible isolates only), *Staphylococcus aureus* (methicillin-susceptible and –resistant isolates), *Streptococcus agalactiae*, *Streptococcus anginosus* grp. (includes *S. anginosus*, *S. intermedius*, and *S. constellatus*), *Streptococcus pyogenes*, *Enterobacter cloacae*, *Klebsiella pneumoniae*, and *Bacteroides fragilis*.

Complicated intra-abdominal infections caused by *Citrobacter freundii*, *Enterobacter cloacae*, *E. coli*, *Klebsiella oxytoca*, *Klebsiella pneumoniae*, *Enterococcus faecalis* (vancomycin-susceptible isolates), *Staphylococcus aureus* (methicillin-susceptible and -resistant isolates only), *Streptococcus anginosus* grp. (includes *S. anginosus*, *S. intermedius*, and *S. constellatus*), *Bacteroides fragilis*, *Bacteroides thetaiotaomicron*, *Bacteroides uniformis*, *Bacteroides vulgatus*, *Clostridium perfringens*, and *Peptostreptococcus micros*.

Appropriate specimens for bacteriological examination should be obtained in order to isolate and identify the causative organisms and to determine their susceptibility to tigecycline. Tigecycline may be initiated as empiric monotherapy before results of these tests are known.

To reduce the development of drug-resistant bacteria and maintain the effectiveness of tigecycline and other antibacterial drugs, tigecycline should be used only to treat infections that are proven or strongly suspected to be caused by susceptible bacteria. When culture and susceptibility information are available, they should be considered in selecting or modifying antibacterial therapy. In the absence of

such data, local epidemiology and susceptibility patterns may contribute to the empiric selection of therapy.

4.2 Posology and method of administration

The recommended dosage regimen for tigecycline is an initial dose of 100 mg, followed by 50 mg every 12 hours. Intravenous (IV) infusions of tigecycline should be administered over approximately 30 to 60 minutes every 12 hours.

The recommended duration of treatment with tigecycline for complicated skin and skin structure infections (cSSSI) or for complicated intra-abdominal infections (cIAI) is 5 to 14 days. The duration of therapy should be guided by the severity and site of the infection and the patient's clinical and bacteriological progress.

Use in patients with renal impairment

No dosage adjustment of tigecycline is necessary in patients with renal impairment or in patients undergoing hemodialysis (see section 5.2).

Use in patients with hepatic impairment

No dosage adjustment is necessary in patients with mild to moderate hepatic impairment (Child-Pugh A and Child Pugh B). Based on the pharmacokinetic profile of tigecycline in patients with severe hepatic impairment (Child Pugh C), the dose of tigecycline should be altered to 100 mg followed by 25 mg every 12 hours. Patients with severe hepatic impairment (Child Pugh C) should be treated with caution and monitored for treatment response (see section 5.2).

Use in children

Clinical trials to establish the safety and effectiveness of tigecycline in patients under 18 years of age have not been conducted. Therefore, use in patients under 18 years of age is not recommended (see section 4.4).

Use in elderly

In a pooled analysis of 3900 subjects who received tigecycline in Phase 3 and 4 clinical studies, 1026 were 65 years and over. Of these, 419 were 75 years and over. No unexpected overall differences in safety were observed between these subjects and younger subjects. No dosage adjustment is necessary in elderly patients.

Race and gender

No dosage adjustment is necessary based on race or gender (See section 5.2).

Mode of administration

Intravenous infusion

4.3 Contraindications

Tigecycline is contraindicated for use in patients who have known hypersensitivity to tigecycline or to any of the excipients listed in section 6.1.

4.4 Special warnings and precautions for use

An increase in all-cause mortality has been observed across Phase 3 and 4 clinical trials in tigecycline-treated subjects versus comparator-treated subjects. In a pooled analysis of all 13 Phase 3 and 4 trials that included a comparator, death occurred in 4.0% (150/3788) of subjects receiving tigecycline and 3.0% (110/3646) of subjects receiving comparator drugs resulting in an unadjusted risk difference of 0.9% (95% CI 0.1, 1.8). In a pooled analysis of these trials, based on a random effects model by trial weight, an adjusted risk difference of all-cause mortality was 0.6% (95% CI 0.1, 1.2) between tigecycline and comparator-treated subjects. The cause of this increase has not been established. This increase in all-cause mortality should be considered when selecting among treatment options (see section 4.8).

Anaphylactic reaction/anaphylactoid reactions have been reported with nearly all antibacterial agents, including tigecycline, and may be life-threatening.

Glycylcycline class antibiotics are structurally similar to tetracycline class antibiotics. Therefore, tigecycline should be administered with caution in patients with known hypersensitivity to tetracycline class antibiotics.

Results of studies in rats with tigecycline have shown bone discoloration. Tigecycline may be associated with permanent tooth discoloration in humans during tooth development.

Pseudomembranous colitis has been reported with nearly all antibacterial agents and may range in severity from mild to life-threatening. Therefore, it is important to consider this diagnosis in patients who present with diarrhoea subsequent to the administration of any antibacterial agent.

Caution should be exercised when considering tigecycline monotherapy in patients with cIAI secondary to clinically apparent intestinal perforation. In Phase 3 and 4 cIAI studies (n=2775), 140/1382 tigecycline-treated subjects and 142/1393 comparator-treated subjects presented with intestinal perforations. Of these subjects, 8/140 subjects treated with tigecycline and 8/142 subjects treated with comparator developed sepsis/septic shock. The relationship of this outcome to treatment cannot be established.

Isolated cases of significant hepatic dysfunction and hepatic failure have been reported in patients being treated with tigecycline.

Glycylcycline class antibiotics are structurally similar to tetracycline class antibiotics and may have similar adverse effects. Such effects may include: photosensitivity, pseudotumor cerebri, pancreatitis and anti-anabolic action (which has led to increased BUN, azotemia, acidosis and hyperphosphatemia).

Pancreatitis acute, which can be fatal, has occurred (frequency: uncommon) in association with tigecycline treatment (see section 4.8). The diagnosis of pancreatitis acute should be considered in patients taking tigecycline who develop clinical symptoms, signs, or laboratory abnormalities suggestive of pancreatitis acute. Cases have been reported in patients without known risk factors for pancreatitis. Patients usually improve after tigecycline discontinuation. Consideration should be given to the cessation of the treatment with tigecycline in patients suspected of having developed pancreatitis.

Monitoring of blood coagulation parameters, including blood fibrinogen, is recommended prior to treatment initiation with tigecycline and regularly while on treatment. (See section 4.8)

The safety and efficacy of tigecycline in patients with hospital acquired pneumonia (HAP) have not been established. In a study of subjects with HAP, subjects were randomized to receive tigecycline (100 mg initially, then 50 mg every 12 hours) or a comparator. In addition, subjects were allowed to receive specified adjunctive therapies. The sub-group of subjects with ventilator-associated pneumonia (VAP) who received tigecycline had lower cure rates (47.9% versus 70.1% for the clinically evaluable population) and greater mortality (25/131 [19.1%] versus 15/122 [12.3%]) than the comparator. Of those subjects with VAP and bacteremia at baseline, those who received tigecycline had greater mortality (9/18 [50.0%] versus 1/13 [7.7%]) than the comparator.

As with other antibiotics preparations, use of this drug may result in overgrowth of non-susceptible organisms, including fungi. Patients should be carefully monitored during therapy. If super infection occurs, appropriate measures should be taken.

4.5 Interaction with other medicinal products and other forms of interaction

Tigecycline (100 mg followed by 50 mg every 12 hours) and digoxin (0.5 mg followed by 0.25 mg every 24 hours) were co-administered to healthy subjects in a drug interaction study. Tigecycline slightly decreased the C_{max} of digoxin by 13%, but did not affect the AUC or clearance of digoxin. This small change in C_{max} did not affect the steady-state pharmacodynamic effects of digoxin as measured by changes in ECG intervals. In addition, digoxin did not affect the pharmacokinetic profile of tigecycline. Therefore, no dosage adjustment is necessary when tigecycline is administered with digoxin.

Concomitant administration of tigecycline (100 mg followed by 50 mg every 12 hours) and warfarin (25 mg single dose) to healthy subjects resulted in a decrease in clearance of R-warfarin and S-warfarin by 40% and 23%, and an increase in AUC by 68% and 29%, respectively. Tigecycline did not significantly alter the effects of warfarin on increased international normalized ratio (INR). In addition, warfarin did not affect the pharmacokinetic profile of tigecycline. However, prothrombin time or other suitable anticoagulation test should be monitored if tigecycline is administered with warfarin.

In vitro studies in human liver microsomes indicate that tigecycline does not inhibit metabolism mediated by any of the following 6 cytochrome CYP450 isoforms: 1A2, 2C8, 2C9, 2C19, 2D6, and 3A4. Therefore, tigecycline is not expected to alter the metabolism of drugs metabolized by these enzymes. In addition, because tigecycline is not extensively metabolized, clearance of tigecycline is not expected to be affected by drugs that inhibit or induce the activity of these CYP450 isoforms.

In vitro studies using Caco-2 cells indicate that tigecycline does not inhibit digoxin flux, suggesting that tigecycline is not a P-glycoprotein (P-gp) inhibitor. This *in vitro* information is consistent with the lack of effect of tigecycline on digoxin clearance noted in the *in vivo* drug interaction study described above.

Tigecycline is a substrate of P-gp based on an *in vitro* study using a cell line overexpressing P-gp. The potential contribution of P-gp-mediated transport to the *in vivo* disposition of tigecycline is not known. Co-administration of P-gp inhibitors (e.g., ketoconazole or cyclosporine) or P-gp inducers (e.g., rifampicin) could affect the pharmacokinetics of tigecycline.

Concurrent use of antibiotics with oral contraceptives may render oral contraceptives less effective.

Concomitant use of tigecycline and calcineurin inhibitors such as tacrolimus or cyclosporine may lead to an increase in serum trough concentrations of the calcineurin inhibitors. Therefore, serum concentrations of the calcineurin inhibitor should be monitored during treatment with tigecycline to avoid drug toxicity.

Interference with laboratory and other diagnostic tests

There are no reported drug-laboratory test interactions.

4.6 Fertility, pregnancy and lactation

Pregnancy

Tigecycline may cause fetal harm when administered to a pregnant woman. Results of animal studies indicate that tigecycline crosses the placenta and is found in fetal tissues. Decreased fetal weights in rats and rabbits (with associated delays in ossification) have been observed with tigecycline.

Tigecycline was not teratogenic in the rat or rabbit (see section 5.3).

There are no adequate and well-controlled studies of tigecycline in pregnant women. Tigecycline should be used during pregnancy only if the potential benefit justifies the potential risk to the fetus.

Tigecycline has not been studied for use during labor and delivery.

Lactation

It is not known whether this drug is excreted in human milk. Available data in animals have shown excretion of tigecycline/metabolites in milk (see section 5.3). Because many drugs are excreted in human milk, caution should be exercised when tigecycline is administered to a nursing woman (see section 4.4).

Fertility

The effects of tigecycline on fertility in humans have not been studied. Nonclinical studies conducted with tigecycline in rats do not indicate harmful effects with respect to fertility or reproductive performance. (see section 5.3).

4.7 Effects on ability to drive and use machines

Tigecycline can cause dizziness (see section 4.8), which may impair the ability to drive and/or operate machinery.

4.8 Undesirable effects

Expected frequency of adverse reactions is presented in CIOMS frequency categories:

Very common	≥10%
Common	≥1% and <10%
Uncommon	≥0.1% and <1%
Rare	≥0.01% and <0.1%
Very rare	<0.01%
Frequency not known	cannot be estimated from the available data

For patients who received tigecycline, the following adverse reactions were reported:

<i>System Organ Class</i>	<i>Adverse Reaction</i>
----------------------------------	--------------------------------

Blood and lymphatic system disorders

Common	Activated partial thromboplastin time prolonged (aPTT), prothrombin time prolonged (PT), thrombocytopenia
Uncommon	International normalised ratio increased (INR)
Rare	Hypofibrinogenaemia

Immune system disorders

Frequency not known	Anaphylactic reaction /anaphylactoid reaction
---------------------	---

Metabolism and nutrition disorders

Common	Hypoproteinaemia, hypoglycaemia, decreased appetite
--------	---

Nervous system disorders

Common	Dizziness, headache
--------	---------------------

Vascular disorders

Common	Phlebitis
Uncommon	Thrombophlebitis

Respiratory, thoracic and mediastinal disorders

Common	Pneumonia
--------	-----------

Gastrointestinal disorders

Very common	Nausea, vomiting, diarrhoea
Common	Abdominal pain, dyspepsia
Uncommon	Pancreatitis acute

Hepatobiliary disorders

Common	Aspartate aminotransferase (AST) increased, alanine aminotransferase (ALT) increased*, hyperbilirubinaemia
Uncommon	Jaundice
Frequency not known	Cholestasis

*AST and ALT abnormalities in tigecycline-treated patients were reported more frequently in the post-therapy period than in those in comparator-treated patients, which occurred more often on therapy.

Skin and subcutaneous tissue disorders*

Common	Pruritus, rash
--------	----------------

Frequency not known Severe skin reactions, including Stevens-Johnson syndrome

General disorders and administration site conditions

Common Impaired healing, injection site reaction
 Uncommon Injection site inflammation, injection site pain, injection site oedema, injection site phlebitis

Investigations

Common Amylase increased, blood urea increased (BUN)

*Tigecycline is structurally similar to tetracycline-class antibacterial drugs and may have similar adverse effects. Such effects may include: Fixed eruption.

In a pooled analysis of all 13 Phase 3 and 4 trials that included a comparator, death occurred in 4.0% (150/3788) of subjects receiving tigecycline and 3.0% (110/3646) of subjects receiving comparator drugs. In a pooled analysis of these trials, the risk difference of all- cause mortality was 0.9% (95% CI 0.1, 1.8) between tigecycline and comparator-treated subjects. In a pooled analysis of these trials, based on a random effects model by trial weight, an adjusted risk difference of all-cause mortality was 0.6% (95% CI 0.1, 1.2) between tigecycline-treated and comparator-treated subjects. No significant differences were observed between tigecycline and comparators within each infection type (see Table 1). The cause of the imbalance has not been established. Generally, deaths were the result of worsening infection, or complications of infection or underlying co-morbidities.

Table 1: Subjects with Outcome of Death by Infection Type					
	Tigecycline		Comparator		Risk Difference*
Infection Type	n/N	%	n/N	%	% (95% CI)
cSSSI	12/834	1.4	6/813	0.7	0.7 (-0.5, 1.9)
cIAI	42/1382	3.0	31/1393	2.2	0.8 (-0.4, 2.1)

cIAI = Complicated intra-abdominal infections; cSSSI = Complicated skin and skin structure infections.

* The difference between the percentage of subjects who died in tigecycline and comparator treatment groups. The 95% CIs were calculated using the Wilson Score Method with continuity correction.

Note: The trials include 300, 305, 900 (cSSSI) and 301, 306, 315, 316, 400 (cIAI).

The most common treatment-emergent adverse reactions in subjects treated with tigecycline were nausea 29.9% (19.3% mild; 9.2% moderate; 1.4% severe) and vomiting 19.9% (12.1% mild; 6.8% moderate; 1.1% severe). In general, nausea or vomiting occurred early (Days 1-2).

Discontinuation from tigecycline was most frequently associated with nausea (1.6%) and vomiting (1.3%).

4.9 Overdose

No specific information is available on the treatment of overdose with tigecycline. Intravenous administration of tigecycline at a single dose of 300 mg over 60 minutes in healthy volunteers resulted in an increased incidence of nausea and vomiting. In single-dose IV toxicity studies conducted with tigecycline in mice, the estimated median lethal dose (LD₅₀) was 124 mg/kg in males and 98 mg/kg in females. In rats, the estimated LD₅₀ was 106 mg/kg for both sexes. Tigecycline is not removed in significant quantities by hemodialysis.

4.10 Abuse and dependence

Drug abuse and dependence have not been demonstrated and are unlikely.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Anti-infective
Glycylcycline antibacterial
ATC code: J01AA12

Mechanism of action

Tigecycline, a glycylcycline antibiotic, inhibits protein translation in bacteria by binding to the 30S ribosomal subunit and blocking entry of amino-acyl tRNA molecules into the A site of the ribosome. This prevents incorporation of amino acid residues into elongating peptide chains. Tigecycline carries a glycyamido moiety attached to the 9-position of minocycline. The substitution pattern is not present in any naturally occurring or semisynthetic tetracycline and imparts certain microbiologic properties that transcend any known tetracycline-derivative *in vitro* or *in vivo* activity. In addition, tigecycline is able to overcome the two major tetracycline resistance mechanisms, ribosomal protection and efflux. However, in recent studies, resistance to tigecycline has been detected in *Enterobacteriales* and other organisms, determined by an efflux pump mechanism and by mutations in a ribosomal protein. Tigecycline has demonstrated *in vitro* and *in vivo* activity against a broad spectrum of bacterial pathogens. There has been no cross resistance observed between tigecycline and other antibiotics. In *in vitro* studies, no antagonism has been observed between tigecycline and other commonly used antibiotics. In general, tigecycline is considered bacteriostatic. At 4 times the minimum inhibitory concentration (MIC), a 2-log reduction in colony counts was observed with tigecycline against *Enterococcus* spp., *Staphylococcus aureus*, and *Escherichia coli*. However, tigecycline has shown some bactericidal activity, and a 3-log reduction was observed against *Neisseria gonorrhoeae*. Tigecycline has also demonstrated bactericidal activity against common respiratory strains of *Streptococcus pneumoniae*, *Haemophilus influenzae*, and *Legionella pneumophila*.

***In Vitro* Susceptibility of Bacteria to Tigecycline**

For broth dilution tests for aerobic organisms, MICs must be determined using testing medium that is fresh (<12 hours old). The disk diffusion procedure utilizes disks impregnated with 15 µg of tigecycline.

EUCAST Reference Information (for markets referencing the EUCAST)

Minimum inhibitory concentration (MIC) and disk inhibition zone breakpoints established by the European Committee on Antimicrobial Susceptibility Testing (EUCAST) are as follows [The European Committee on Antimicrobial Susceptibility Testing]. Breakpoint tables for interpretation of MICs and zone diameters:

Table 2. EUCAST Breakpoints		
Pathogen	MIC (mg/L)	Inhibition zone diameter (mm)
	≤S (Susceptible)/ >R (Resistant)	≥S (Susceptible)/ <R (Resistant)
<i>Enterobacterales</i> (formerly <i>Enterobacteriaceae</i>): <i>Escherichia coli</i> and <i>Citrobacter koseri</i> : ^(†)	≤ 0.5 / > 0.5	≥ 18 / < 18 ^(*)
<i>Staphylococcus</i> spp.	≤ 0.5 / > 0.5	≥ 19 / < 19
<i>Enterococcus faecalis</i>	≤ 0.25 / > 0.25	≥ 20 / < 20
<i>Enterococcus faecium</i>	≤ 0.25 / > 0.25	≥ 22 / < 22
<i>Streptococcus</i> groups A, B, C and G	≤ 0.125 / > 0.125	≥ 19 / < 19
PK/PD (non-species related)		
	≤ 0.5 / > 0.5	-
^(†) For other <i>Enterobacterales</i> , the activity of tigecycline varies from insufficient in <i>Proteus</i> spp., <i>Morganella morganii</i> and <i>Providencia</i> spp. to variable in other species. ^(*) Zone diameter breakpoints validated for <i>E. coli</i> only. For <i>C. koseri</i> use MIC method.		

For anaerobic bacteria there is clinical evidence of efficacy in polymicrobial intra-abdominal infections, but no correlation between MIC values, PK/PD data and clinical outcome. Therefore, no breakpoint for susceptibility is given.

Quality control ranges for EUCAST susceptibility testing are in the following table.

Table 3: Acceptable Quality Control Ranges for Susceptibility Testing (EUCAST)		
Organism	MIC range (mg/L)	Inhibition zone diameter range (mm)
<i>Escherichia coli</i> ATCC 25922	0.03-0.25	20-27
<i>Staphylococcus aureus</i> ATCC 29213	0.03-0.25	19-25
<i>Enterococcus faecalis</i> ATCC 29212	0.03-0.125	20-26
<i>Streptococcus pneumoniae</i> ATCC 49619	0.016-0.125	24-30
ATCC = American Type Culture Collection.		

PK/PD relationship

Limited animal data indicates that AUC/MIC is the pharmacodynamic index best related to outcome. Human pharmacodynamic studies indicate a relationship between AUC/MIC and clinical as well microbiological efficacy.

Susceptibility

The prevalence of acquired resistance may vary geographically and with time for selected species, and local information on resistance is desirable, particularly when treating severe infections.

The information below provides only approximate guidance on the probability as to whether the microorganism will be susceptible to tigecycline or not.

Pathogen
Commonly Susceptible Species <u>Gram-positive Aerobes</u> <i>Enterococcus</i> spp.† (including vancomycin resistant isolates) <i>Listeria monocytogenes</i> <i>Staphylococcus aureus</i>* (including methicillin resistant isolates) <i>Staphylococcus epidermidis</i> (including methicillin resistant isolates) <i>Staphylococcus haemolyticus</i> <i>Streptococcus agalactiae</i>* <i>Streptococcus pyogenes</i>* <i>Streptococcus pneumoniae</i>† Viridans group streptococci† <u>Gram-negative Aerobes</u> <i>Aeromonas hydrophila</i> <i>Citrobacter freundii</i>* <i>Citrobacter koseri</i> <i>Escherichia coli</i>* <i>Haemophilus influenzae</i>* <i>Legionella pneumophila</i>* <i>Moraxella catarrhalis</i>* <i>Neisseria gonorrhoeae</i> <i>Neisseria meningitidis</i> <i>Pasteurella multocida</i> <u>Anaerobes</u> <i>Clostridioides difficile</i> <i>Clostridium perfringens</i>* <i>Peptostreptococcus</i> spp. † <i>Porphyromonas</i> spp. <i>Prevotella</i> spp. <u>Other organisms</u> <i>Chlamydia pneumoniae</i> <i>Mycobacterium abscessus</i> <i>Mycobacterium chelonae</i> <i>Mycobacterium fortuitum</i> <i>Mycoplasma pneumoniae</i>
Species for which acquired resistance may be a problem <u>Gram-negative Aerobes</u> <i>Acinetobacter baumannii</i> <i>Enterobacter cloacae</i>* <i>Klebsiella aerogenes</i> <i>Klebsiella oxytoca</i>* <i>Klebsiella pneumoniae</i>*

Pathogen
<i>Morganella morganii</i> <i>Salmonella</i> spp <i>Serratia marcescens</i> <i>Shigella</i> spp <i>Stenotrophomonas maltophilia</i> <u>Anaerobes</u> <i>Bacteroides fragilis</i> group† <i>Parabacteroides distasonis</i>
Inherently resistant organisms
<u>Gram-negative Aerobes</u> <i>Providencia</i> spp <i>Proteus</i> spp. <i>Pseudomonas aeruginosa</i>
*denotes species against which it is considered that activity has been satisfactorily demonstrated in clinical studies. †activity in clinical studies has been demonstrated for vancomycin-susceptible <i>Enterococcus faecalis</i>; penicillin-susceptible pneumococci; among viridans streptococci for the <i>Streptococcus anginosus</i> group (includes <i>S. anginosus</i>, <i>S. intermedius</i> and <i>S. constellatus</i>); among <i>Peptostreptococcus</i> spp. for <i>P. micros</i>; among <i>Bacteroides</i> spp. for <i>B. fragilis</i>, <i>B. thetaiotaomicron</i>, <i>B. uniformis</i>, <i>B. ovatus</i> and <i>B. vulgatus</i>.

Clinical trial data on Efficacy

Complicated Skin and Skin Structure Infections (cSSSI)

Tigecycline was evaluated in adults for the treatment of (cSSSI) in two randomized, double-blind, active-controlled, multinational, multicenter studies. These studies compared tigecycline (100 mg IV initial dose followed by 50 mg every 12 hours) with vancomycin (1g IV every 12 hours)/aztreonam (2g IV every 12 hours) for 5 to 14 days. Subjects with complicated deep soft tissue infections, including wound infections and cellulitis (≥ 10 cm, requiring surgery/drainage or with complicated, underlying disease), major abscesses, infected ulcers, and burns were enrolled in the studies. The primary efficacy endpoint was the clinical response at the test of cure (TOC) visit in the co-primary populations of the clinically evaluable (CE) and clinical modified intent-to-treat (c-mITT) subjects. See Table 4.

Table 4: Clinical Cure Rates from Two Pivotal Studies in cSSSI after 5 to 14 Days of Therapy		
	Tigecycline ^a	Vancomycin/ Aztreonam ^b
	n/N (%)	n/N (%)
CE	365/422 (86.5)	364/411 (88.6)
c-mITT	429/538 (79.7)	425/519 (81.9)
^a 100 mg initially, followed by 50 mg every 12 hours.		
^b Vancomycin (1 g IV every 12 hours)/Aztreonam (2 g IV every 12 hours).		

Clinical cure rates at TOC by pathogen in microbiologically evaluable (ME) subjects with cSSSI are presented in the Table 5.

Table 5. Clinical Cure Rates by Infecting Pathogen in ME Subjects with cSSSI^a		
Pathogen	Tigecycline n/N (%)	Vancomycin/Aztreonam n/N (%)
<i>Escherichia coli</i>	29/36 (80.6)	26/30 (86.7)
<i>Enterobacter cloacae</i>	10/12 (83.3)	15/15 (100)
<i>Enterococcus faecalis</i> (vancomycin-susceptible only)	15/21 (71.4)	19/24 (79.2)
Methicillin-susceptible <i>Staphylococcus aureus</i> (MSSA) ^b	124/137 (90.5)	113/120 (94.2)
Methicillin-resistant <i>Staphylococcus aureus</i> (MRSA) ^b	79/95 (83.2)	46/57 (80.7)
<i>Streptococcus agalactiae</i>	8/8 (100)	11/14 (78.6)
<i>Streptococcus anginosus</i> grp. ^c	17/21 (81.0)	9/10 (90.0)
<i>Streptococcus pyogenes</i>	31/32 (96.9)	24/27 (88.9)
<i>Bacteroides fragilis</i>	7/9 (77.8)	4/5 (80.0)
^a Two pivotal studies from cSSSI and two Phase 3 Resistant Pathogen studies		
^b Includes cases of concurrent bacteremia		
^c Includes <i>Streptococcus anginosus</i> , <i>Streptococcus intermedius</i> , and <i>Streptococcus constellatus</i>		

Complicated Intra-abdominal Infections (cIAI)

Tigecycline was evaluated in adults for the treatment of cIAI in two randomized, double-blind, active-controlled, multinational, multicenter studies. These studies compared tigecycline (100 mg IV initial dose followed by 50 mg every 12 hours) with imipenem/cilastatin (500 mg IV every 6 hours) for 5 to 14 days. Subjects with complicated diagnoses including appendicitis, cholecystitis, diverticulitis, gastric/duodenal perforation, intra-abdominal abscess, perforation of the intestine, and peritonitis were enrolled in the studies. The primary efficacy endpoint was the clinical response at the TOC visit for the co-primary populations of the ME and the microbiologic modified intent-to-treat (m-mITT) subjects. See Table 6.

Table 6: Clinical Cure Rates from Two Pivotal Studies in cIAI		
	Tigecycline ^a n/N (%)	Imipenem/Cilastatin ^b n/N (%)
ME	441/512 (86.1)	442/513 (86.2)
m-mITT	506/631 (80.2)	514/631 (81.5)
^a 100 mg initially, followed by 50 mg every 12 hours.		
^b Imipenem/Cilastatin (500 mg every 6 hours).		

Clinical cure rates at TOC by pathogen in the ME subjects with cIAI are presented in Table 7.

Table 7: Clinical Cure Rates by Infecting Pathogen in ME Subjects with cIAI^a		
Pathogen	Tigecycline n/N (%)	Imipenem/ Cilastatin n/N (%)
<i>Citrobacter freundii</i>	12/16 (75.0)	3/4 (75.0)
<i>Enterobacter cloacae</i>	15/17 (88.2)	16/17 (94.1)
<i>Escherichia coli</i>	284/336 (84.5)	297/342 (86.8)
<i>Klebsiella oxytoca</i>	19/20 (95.0)	17/19 (89.5)
<i>Klebsiella pneumoniae</i> ^b	42/47 (89.4)	46/53 (86.8)
<i>Enterococcus faecalis</i>	29/38 (76.3)	35/47 (74.5)
Methicillin-susceptible	26/28 (92.9)	22/24 (91.7)
<i>Staphylococcus aureus</i> (MSSA) ^c		
Methicillin-resistant	16/18 (88.9)	1/3 (33.3)
<i>Staphylococcus aureus</i> (MRSA) ^c		
<i>Streptococcus anginosus</i> grp. ^d	101/119 (84.9)	60/79 (75.9)
<i>Bacteroides fragilis</i>	68/88 (77.3)	59/73 (80.8)
<i>Bacteroides thetaiotaomicron</i>	36/41 (87.8)	31/36 (86.1)
<i>Bacteroides uniformis</i>	12/17 (70.6)	14/16 (87.5)
<i>Bacteroides vulgatus</i>	14/16 (87.5)	4/6 (66.7)
<i>Clostridium perfringens</i>	18/19 (94.7)	20/22 (90.9)
<i>Peptostreptococcus micros</i>	13/17 (76.5)	8/11 (72.7)
^a Two cIAI pivotal studies and two Phase 3 Resistant Pathogen studies. ^b Includes ESBL producing isolates. ^c Includes cases of concurrent bacteremia. ^d Includes <i>Streptococcus anginosus</i> , <i>Streptococcus intermedius</i> , and <i>Streptococcus constellatus</i> .		

Methicillin-Resistant *Staphylococcus aureus* (MRSA)

Tigecycline was evaluated in adults for the treatment of various serious infections (cIAI, cSSSI, and other infections) due to MRSA in Study 307.

Study 307 was a randomized, double-blind, active-controlled, multinational, multicenter study evaluating tigecycline (100 mg IV initial dose followed by 50 mg every 12 hours) and vancomycin (1 g IV every 12 hours) for the treatment of infections due to MRSA. Subjects with cIAI, cSSSI, and other infections were enrolled in this study. The primary efficacy endpoint was the clinical response at the TOC visit for the co-primary populations of the ME and the m-mITT subjects. For clinical cure rates see Table 8 for MRSA.

Table 8. Clinical Cure Rates from Resistant Pathogen Study 307^a for MRSA after 7 to 28 Days of Therapy		
	Tigecycline ^b n/N (%)	Vancomycin ^c n/N (%)
Study 307		
ME	70/86 (81.4)	26/31 (83.9)
cIAI	13/14 (92.9)	4/4 (100.0)
cSSSI	51/59 (86.4)	20/23 (87.0)
m-mITT	75/100 (75.0)	27/33 (81.8)
cIAI	13/15 (86.7)	5/6 (83.3)

Table 8. Clinical Cure Rates from Resistant Pathogen Study 307^a for MRSA after 7 to 28 Days of Therapy

	Tigecycline ^b n/N (%)	Vancomycin ^c n/N (%)
cSSSI	55/70 (78.6)	20/23 (87.0)
^a Study included subjects with cIAI, cSSSI, and other infections. ^b 100 mg initially, followed by 50 mg every 12 hours. ^c 1 g IV every 12 hours.		

Cardiac Electrophysiology

No significant effect of a single intravenous dose of tigecycline 50 mg or 200 mg on QTc interval was detected in a randomized, placebo- and active-controlled four-arm crossover thorough QTc study of 46 healthy subjects.

5.2 Pharmacokinetic properties

The mean pharmacokinetic parameters of tigecycline for the recommended dosage regimen after single and multiple intravenous doses are summarized in Table 9.

Intravenous infusions of tigecycline should be administered over approximately 30 to 60 minutes.

Table 9: Mean (CV%) Pharmacokinetic Parameters of Tigecycline

	Single Dose 100 mg	Multiple Dose ^c 50 mg q12h
C _{max} (µg/mL) ^a	1.45 (22%)	0.87 (27%)
C _{max} (µg/mL) ^b	0.90 (30%)	0.63 (15%)
AUC (µg·h/mL)	5.19 (36%)	-
AUC _{0-24h} (µg·h/mL)	-	4.70 (36%)
C _{min} (µg/mL)	-	0.13 (59%)
t _{1/2} (h)	27.1 (53%)	42.4 (83%)
CL (L/h)	21.8 (40%)	23.8 (33%)
CL _r (mL/min)	38.0 (82%)	51.0 (58%)
V _{ss} (L)	568 (43%)	639 (48%)
^a 30-minute infusion. ^b 60-minute infusion. ^c 100 mg initially, followed by 50 mg every 12 hours.		

Absorption

Tigecycline is administered intravenously, and therefore has 100% bioavailability.

Distribution

The *in vitro* plasma protein binding of tigecycline ranges from approximately 71% to 89% at concentrations observed in clinical studies (0.1 to 1.0 µg/mL). Animal and human pharmacokinetic studies have demonstrated that tigecycline readily distributes to tissues. In rats receiving single or multiple doses of ¹⁴C-tigecycline, radioactivity was well distributed to most tissues, with the highest overall exposure observed in bone, bone marrow, thyroid gland, kidney, spleen, and salivary gland. In humans, the steady-state volume of distribution of tigecycline averaged 500 to 700 L (7 to 9 L/kg), indicating tigecycline is extensively distributed beyond the plasma volume and into the tissues of humans.

Two studies examined the steady-state pharmacokinetic profile of tigecycline in specific tissues or fluids of healthy subjects receiving tigecycline 100 mg followed by 50 mg every 12 hours. In a bronchoalveolar lavage study, the tigecycline AUC_{0-12h} (134 µg·h/mL) in alveolar cells was approximately 77.5-fold higher than the AUC_{0-12h} in the serum of these subjects, and the AUC_{0-12h} (2.28 µg·h/mL) in epithelial lining fluid was approximately 32% higher than the AUC_{0-12h} in serum. In a skin blister study, the AUC_{0-12h} (1.61 µg·h/mL) of tigecycline in skin blister fluid was approximately 26% lower than the AUC_{0-12h} in the serum of these subjects.

In a single-dose study, tigecycline 100 mg was administered to subjects prior to undergoing elective surgery or medical procedure for tissue extraction. Tissue concentrations at 4 hours after tigecycline administration were measured in the following tissue and fluid samples: gallbladder, lung, colon, synovial fluid, and bone. Tigecycline attained higher concentrations in tissues versus serum in gallbladder (38-fold, n=6), lung (3.7-fold, n=5), and colon (2.3-fold, n=6). The concentration of tigecycline in these tissues after multiple doses has not been studied.

Metabolism

Tigecycline is not extensively metabolized. *In vitro* studies with tigecycline using human liver microsomes, liver slices, and hepatocytes led to the formation of only trace amounts of metabolites. In healthy male volunteers receiving ¹⁴C-tigecycline, tigecycline was the primary ¹⁴C-labeled material recovered in urine and feces, but a glucuronide, an N-acetyl metabolite, and a tigecycline epimer (each at no more than 10% of the administered dose) were also present.

Elimination

The recovery of total radioactivity in feces and urine following administration of ¹⁴C-tigecycline indicates that 59% of the dose is eliminated by biliary/fecal excretion, and 33% is excreted in urine. Overall, the primary route of elimination for tigecycline is biliary excretion of unchanged tigecycline. Glucuronidation and renal excretion of unchanged tigecycline are secondary routes.

Tigecycline is a substrate of P-gp based on an *in vitro* study using a cell line overexpressing P-gp. The potential contribution of P-gp-mediated transport to the *in vivo* disposition of tigecycline is not known.

Special Populations

Hepatic insufficiency

In a study comparing 10 subjects with mild hepatic impairment (Child Pugh A), 10 subjects with moderate hepatic impairment (Child Pugh B), and five subjects with severe hepatic impairment (Child Pugh C) to 23 age and weight matched healthy control subjects, the single-dose pharmacokinetic

disposition of tigecycline was not altered in subjects with mild hepatic impairment. However, systemic clearance of tigecycline was reduced by 25% and the half-life of tigecycline was prolonged by 23% in subjects with moderate hepatic impairment (Child Pugh B). In addition, systemic clearance of tigecycline was reduced by 55%, and the half-life of tigecycline was prolonged by 43% in subjects with severe hepatic impairment (Child Pugh C).

Based on the pharmacokinetic profile of tigecycline, no dosage adjustment is warranted in subjects with mild to moderate hepatic impairment (Child Pugh A and Child Pugh B). However, in subjects with severe hepatic impairment (Child Pugh C), the dose of tigecycline should be reduced to 100 mg followed by 25 mg every 12 hours. Subjects with severe hepatic impairment (Child Pugh C) should be treated with caution and monitored for treatment response (see section 4.2).

Renal insufficiency

A single-dose study compared 6 subjects with severe renal impairment (creatinine clearance $Cl_{Cr} \leq 30$ mL/min), 4 end stage renal disease subjects receiving tigecycline 2 hours before hemodialysis, 4 end stage renal disease subjects receiving tigecycline after hemodialysis, and 6 healthy control subjects. The pharmacokinetic profile of tigecycline was not altered in any of the renally-impaired subject groups, nor was tigecycline removed by hemodialysis. No dosage adjustment of tigecycline is necessary in subjects with renal impairment or in subjects undergoing hemodialysis (see section 4.2).

Elderly

No overall differences in pharmacokinetics were observed between healthy elderly subjects ($n = 15$, age 65 – 75; $n = 13$, age > 75 and younger subjects ($n = 18$) receiving a single, 100 mg dose of tigecycline. Therefore, no dosage adjustment is necessary based on age.

Children

The pharmacokinetics of tigecycline in patients less than 18 years of age have not been established.

Gender

In a pooled analysis of 38 women and 298 men participating in clinical pharmacology studies, there was no significant difference in the mean ($\pm$ SD) tigecycline clearance between women (20.7 ± 6.5 L/h) and men (22.8 ± 8.7 L/h). Therefore, no dosage adjustment is necessary based on gender.

Race

In a pooled analysis of 73 Asian subjects, 53 black subjects, 15 Hispanic subjects, 190 white subjects, and 3 subjects classified as “other” participating in clinical pharmacology studies, there was no significant difference in the mean ($\pm$ SD) tigecycline clearance among the Asian subjects (28.8 ± 8.8 L/h), black subjects (23.0 ± 7.8 L/h), Hispanic subjects (24.3 ± 6.5 L/h), white subjects (22.1 ± 8.9 L/h), and “other” subjects (25.0 ± 4.8 L/h). Therefore, no dosage adjustment is necessary based on race.

5.3 Preclinical safety data

Carcinogenicity

Lifetime studies in animals have not been performed to evaluate the carcinogenic potential of tigecycline.

Mutagenicity

No mutagenic or clastogenic potential was found in a battery of tests, including an *in vitro* chromosome aberration assay in Chinese hamster ovary (CHO) cells, *in vitro* forward mutation assay in CHO cells (HGRPT locus), *in vitro* forward mutation assays in mouse lymphoma cells, and *in vivo* micronucleus assay.

Reproduction toxicity

Tigecycline did not affect mating or fertility in rats at exposures up to 4.7 times the human daily dose based on AUC. In female rats, there were no compound-related effects on ovaries or estrus cycles at exposures up to 4.7 times the human daily dose based on AUC.

In preclinical safety studies, ¹⁴C-labeled tigecycline crossed the placenta and was found in fetal tissues, including fetal bony structures. The administration of tigecycline was associated with slight reductions in fetal weights and an increased incidence of minor skeletal anomalies (delays in bone ossification) at exposures of 4.7 times and 1.1 times the human daily dose based on AUC in rats and rabbits, respectively.

Results from animal studies using ¹⁴C-labeled tigecycline indicate that tigecycline is excreted readily via the milk of lactating rats. Consistent with the limited oral bioavailability of tigecycline, there is little or no systemic exposure to tigecycline in the nursing pups as a result of exposure via the maternal milk.

Other

Decreased erythrocytes, reticulocytes, leukocytes and platelets, in association with bone marrow hypocellularity, have been seen with tigecycline at exposures of 8.1 times and 9.8 times the human daily dose based on AUC in rats and dogs, respectively. These alterations were shown to be reversible after two weeks of dosing.

Bolus intravenous administration of tigecycline has been associated with a histamine response in preclinical studies. These effects were observed at exposures of 14.3 and 2.8 times the human daily dose based on the AUC in rats and dogs, respectively.

No evidence of photosensitivity was observed in rats following administration of tigecycline.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Lactose monohydrate, hydrochloric acid, sodium hydroxide and nitrogen.

6.2 Incompatibilities

Compatible intravenous solutions include 0.9% Sodium Chloride Injection, USP, 5% Dextrose Injection, USP, and Lactated Ringer's Injection, USP.

Tigecycline is compatible with the following drugs or diluents when used with either 0.9% Sodium Chloride Injection, USP, or 5% Dextrose Injection, USP and administered simultaneously through the same line: amikacin, dobutamine, dopamine HCl, gentamicin, haloperidol, Lactated Ringer's, lidocaine HCl, metoclopramide, morphine, norepinephrine, piperacillin/tazobactam (EDTA formulation), potassium chloride, propofol, ranitidine HCl, theophylline, and tobramycin.

The following drugs should not be administered simultaneously through the same line as tigecycline: amphotericin B, amphotericin B lipid complex, diazepam, esomeprazole and omeprazole.

6.3 Shelf-life

Observe “Expiry date” (month/year) imprinted on outer carton.

6.4 Special precautions for storage

Store below 30°C.

Once reconstituted, tigecycline should be used immediately.

Reconstituted solution must be transferred and further diluted for IV infusion.

6.5 Nature and contents of container

Tigecycline for injection is supplied in a single-dose 5 mL glass vial containing 50 mg lyophilized powder for reconstitution.

Supplied 10 vials/box.

6.6 Special precautions for disposal and other handling

The lyophilized powder should be reconstituted with 5.3 mL of 0.9% Sodium Chloride Injection, USP, or 5% Dextrose Injection, USP, to achieve a concentration of 10 mg/mL of tigecycline. The vial should be gently swirled until the drug dissolves. Withdraw 5 mL of the reconstituted solution from the vial and add to a 100 mL IV bag for infusion. For a 100 mg dose, reconstitute using two vials into a 100 mL IV bag (Note: The vial contains a 6% overage. Thus, 5 mL of reconstituted solution is equivalent to 50 mg of the drug). The reconstituted solution should be yellow to orange in colour; if not, the solution should be discarded. Parenteral drug products should be inspected visually for particulate matter and discoloration (e.g., green or black) prior to administration whenever solution and container permit. Once reconstituted, tigecycline should be used immediately.

Tigecycline may be administered intravenously through a dedicated line or through a Y-site. If the same intravenous line is used for sequential infusion of several drugs, the line should be flushed before and after infusion of tigecycline with either 0.9% Sodium Chloride Injection, USP, or 5% Dextrose Injection, USP. Injection should be made with an infusion solution compatible with tigecycline and with any other drug(s) administered via this common line (see section 6.2).

7. MANUFACTURER

Wyeth Lederle S.r.l
Via Franco Gorgone
Zona Industriale
Catania CT, Italy

Date of Revision: 19 MAR 2025

TYGACIL - 0325